

CLOUD COMPUTING COURSE

Lecture3-Cloud Computing Models

ABSTRACT
Lecture3 Introduces cloud computing models.

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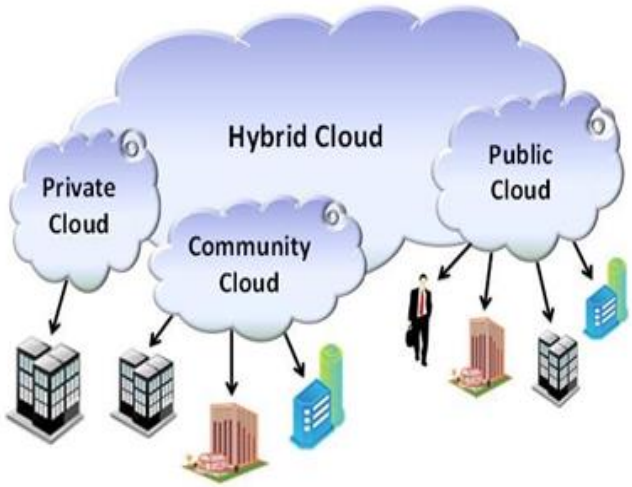
Cloud Computing Models

There are certain services and models working behind the scene making the cloud computing feasible and accessible to end users. Following are the working models for cloud computing:

- Deployment Models
- Service Models

Cloud Computing Deployment Models

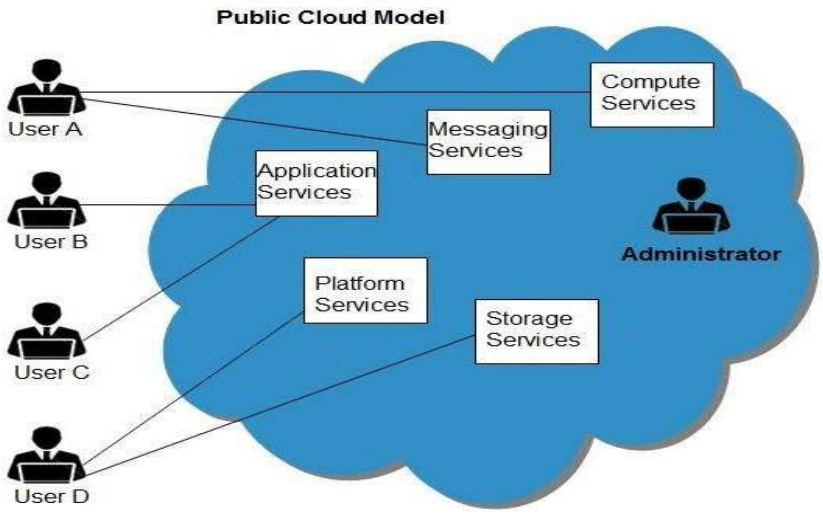
Deployment models define the type of access to the cloud, i.e., how the cloud is located? Cloud can have any of the four types of access: Public, Private, Hybrid, and Community.



1. Public Cloud

Public Cloud allows systems and services to be easily accessible to public.

The IT giants such as **Google**, **Amazon** and **Microsoft** offer cloud services via Internet.

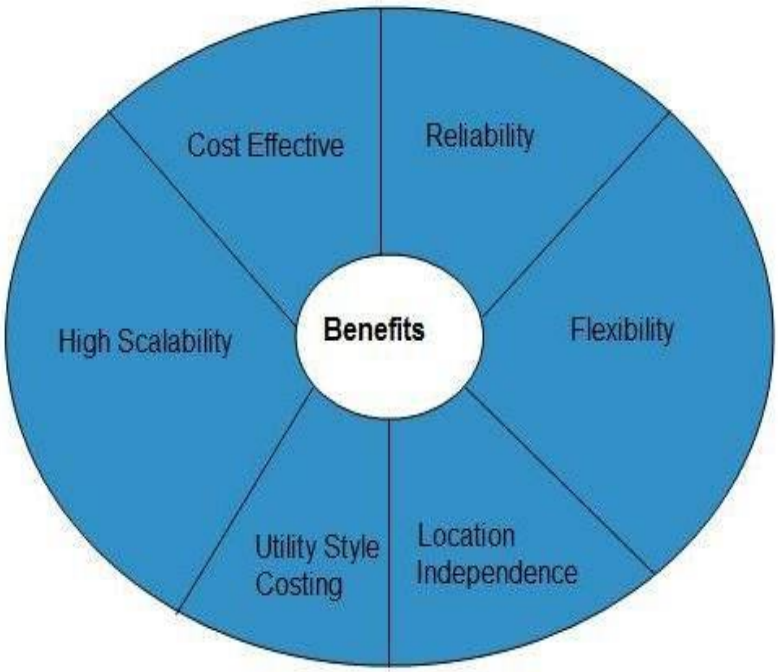


Public Cloud Model :

The Public Cloud Model is shown in the diagram next.

Public Cloud Advantages

There are many benefits of deploying cloud as public cloud model. The following diagram shows some of those benefits:



Cost Effective

Since public cloud shares same resources with large number of customers, it turns out inexpensive.

Reliability

The public cloud employs large number of resources from different locations. If any of the resources fails, public cloud can employ another one.

Flexibility

The public cloud can smoothly integrate with private cloud, which gives customers a flexible approach.

Location Independence

Public cloud services are delivered through Internet, ensuring location independence.

Utility Style Costing

Public cloud is also based on pay-per-use model and resources are accessible whenever customer needs them.

High Scalability

Cloud resources are made available on demand from a pool of resources, i.e., they can be scaled up or down according the requirement.

Public Cloud Disadvantages

Low Security: Data is hosted off-site and resources are shared publicly.

Less Customizable: It is comparatively less customizable than private cloud.

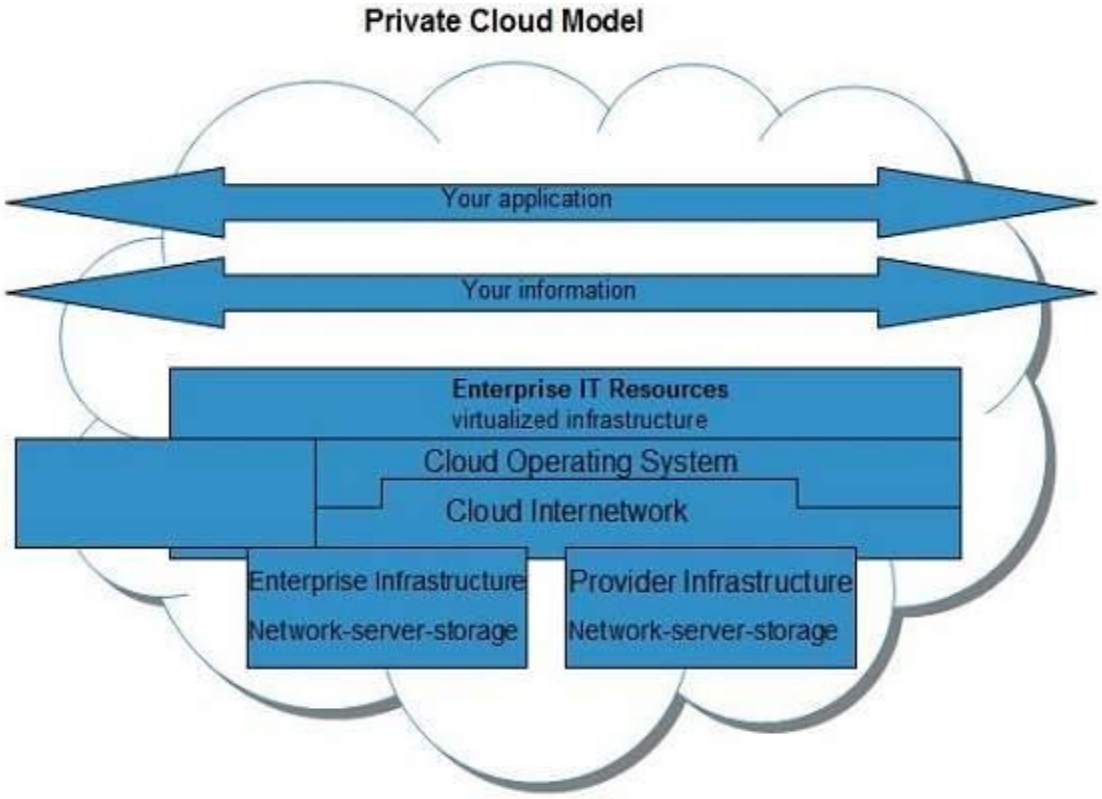
2. Private Cloud

Private Cloud allows systems and services to be accessible within an organization. The Private Cloud is operated only within a single organization. However, it managed internally by the organization itself or by third-party.

Private cloud providers such as **Microsoft**, **VMware**

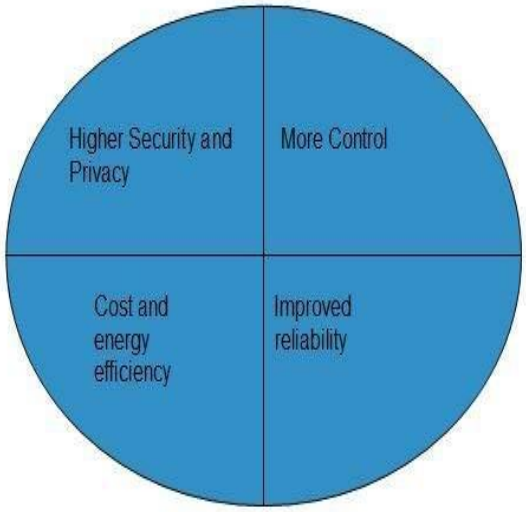
Private Cloud Model

The private cloud model is shown in the diagram below.



Private Cloud Advantages

There are many benefits of deploying cloud as private cloud model. The following diagram shows some of those benefits:



High Security and Privacy

Private cloud operations are not available to general public and resources are shared from distinct pool of resources. Therefore, it ensures high **security** and **privacy.**

More Control

The private cloud has more control on its resources and hardware than public cloud because it is accessed only within an organization.

Cost and Energy Efficiency

The **private cloud** resources are not as cost effective as resources in public clouds but they offer more efficiency than public cloud resources.

Private Cloud Disadvantages

Here are the disadvantages of using private cloud model:

Restricted Area of Operation

The private cloud is only accessible locally and is very difficult to deploy globally.

High Priced

Purchasing new hardware in order to fulfill the demand is a costly transaction.

Limited Scalability

The private cloud can be scaled only within capacity of internal hosted resources.

Additional Skills

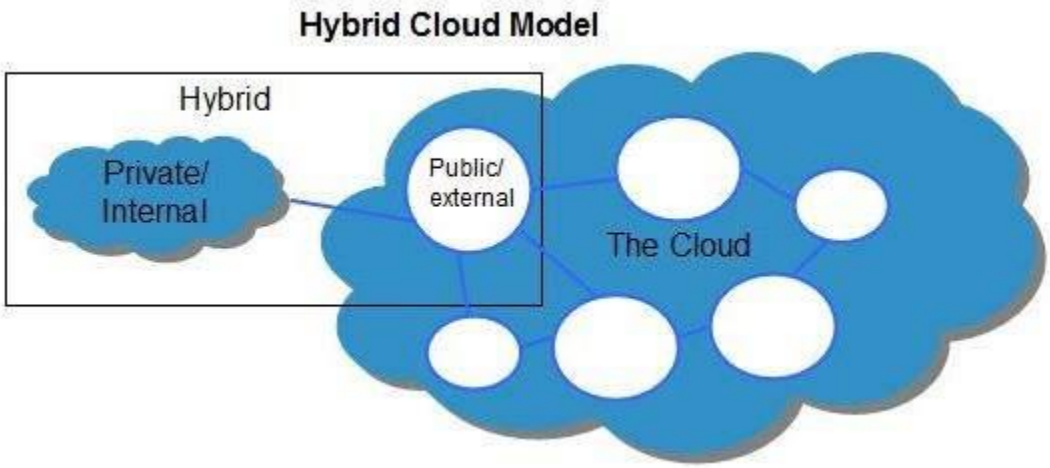
In order to maintain cloud deployment, organization requires skilled expertise.

3. Hybrid Cloud

The hybrid cloud is a mixture of public and private cloud, in which the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

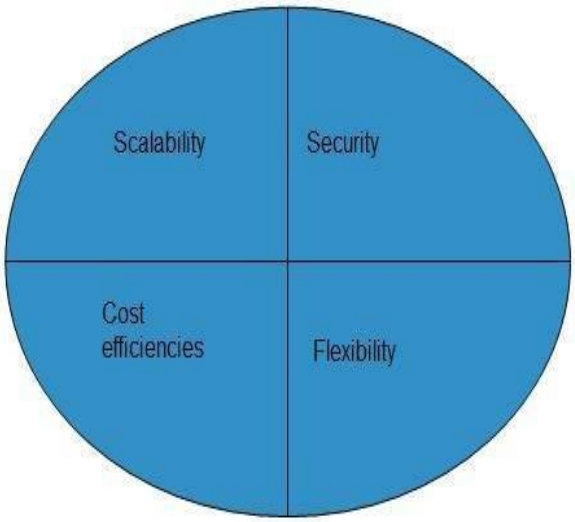
Hybrid Cloud is a mixture of public and private cloud. Non-critical activities are performed using public cloud while the critical activities are performed using private cloud.

Hybrid Cloud Model



Hybrid Cloud Advantage

There are many benefits of deploying cloud as hybrid cloud model. The following diagram shows some of those benefits:



Scalability

It offers features of both, the public cloud scalability and the private cloud scalability.

Flexibility

It offers secure resources and scalable public resources.

Cost Efficiency

Public clouds are more cost effective than private ones. Therefore, hybrid clouds can be cost saving.

Security

The private cloud in hybrid cloud ensures higher degree of security.

Hybrid Cloud Disadvantages

Networking Issues

Networking becomes complex due to presence of private and public cloud.

Security Compliance

It is necessary to ensure that cloud services are compliant with security policies of the organization.

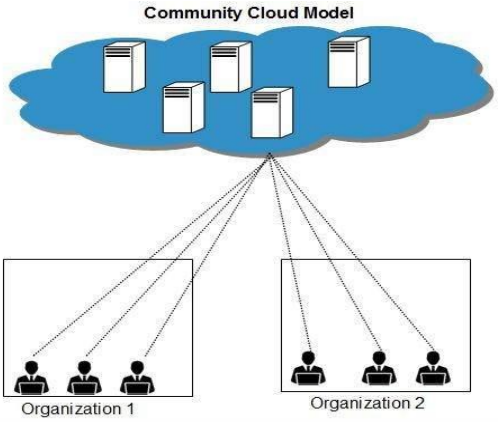
Infrastructure Dependency

The hybrid cloud model is dependent on internal IT infrastructure, therefore it is necessary to ensure redundancy across data centers.

4. Community Cloud

It allows system and services to be accessible by group of organizations. It shares the infrastructure between several organizations from a specific community. It may be managed internally by organizations or by the third-party.

Community Cloud Model



Community Cloud Advantages



There are many benefits of deploying cloud as community cloud model.

Cost Effective

Community cloud offers same advantages as that of private cloud at low cost.

Sharing Among Organizations

Community cloud provides an infrastructure to share cloud resources and capabilities among several organizations.

Security

The community cloud is comparatively more secure than the public cloud but less secured than the private cloud.

Issues

- Since all data is located at one place, one must be careful in storing data in community cloud because it might be accessible to others.
- It is also challenging to allocate responsibilities of governance, security and cost among organizations.